

DropBox

Functional requirements:

user should be able to upload/download any file based on his preference,
user should be able to search for any file among the existing files based on his preference,
user should be able to sync up all the uploads & downloads over app

Non-functional requirements :

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem(Here eventual consistency is fine , but here we need 999.99% uptime i.e
Availability>>>Consistency)

Core Entities:

user ,
file,
filemetadata,

API :

POST:/upload

```
{  
  userID,  
  {FileMetadata}  
}
```

GET:/search{fileID}

HLD :

